* One hot Encoding

lets take example where

Color	Target
yellow	0
red	1
Blue	1
yellow	0
green	1
red	0
red	1

now we will not assign [0,1,2,3] here as every element in that variable is equal, instead we create new columns for each category value.

Color_y	Color_r	Color_B	Color_G	Target
1	0	0	0	0
0	1	0	0	1
0	0	1	0	1
1	0	0	0	0
0	0	0	1	1
0	1	0	0	0
0	1	0	0	1

So after encoding we remove one column. because

We cannot let have a mathematical relationship between dummy variable (new column created after encoding) or multicollinearity can occur. This can give problem in linear column.

avoid

That is to Dummy Variable Trap.

In the case of data set where in a category there are many categories, there data we keep n most frequent categories column and then very less used categories will be put in a single category like other.

Its like instead of dealing 50 category column we will deal with 10 category column.

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